

REGIONAL SALES PERFORMANCE ANALYSIS

DATA STORYTELLING & STATISTICAL VALIDATION

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Internship Project



Business Objective

The objective of this project is to analyze regional sales performance and identify important business trends across revenue generation, product categories, customer payment behavior, and monthly sales performance.

The analysis aims to support data-driven business decision-making through business intelligence and statistical validation.



Dataset Overview

Key Variables:

- Total Revenue
- Units Sold
- Product Category
- Region
- Payment Method
- Monthly Sales Data

240

Total Records

14

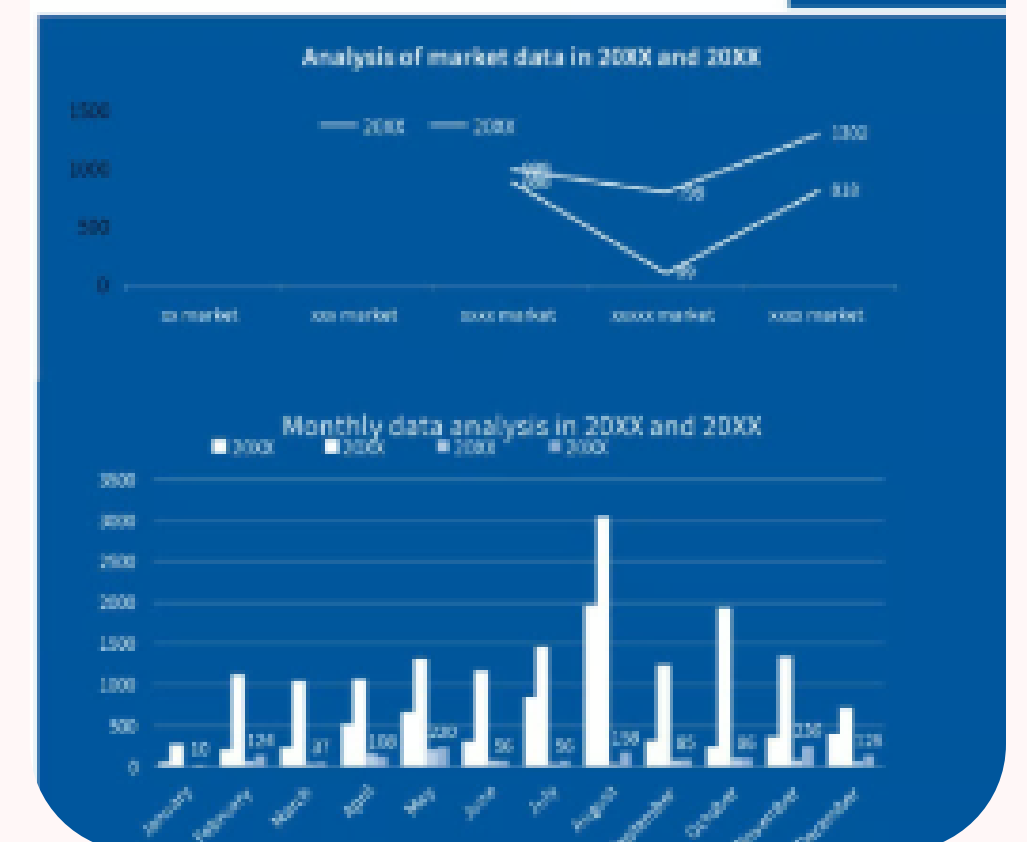
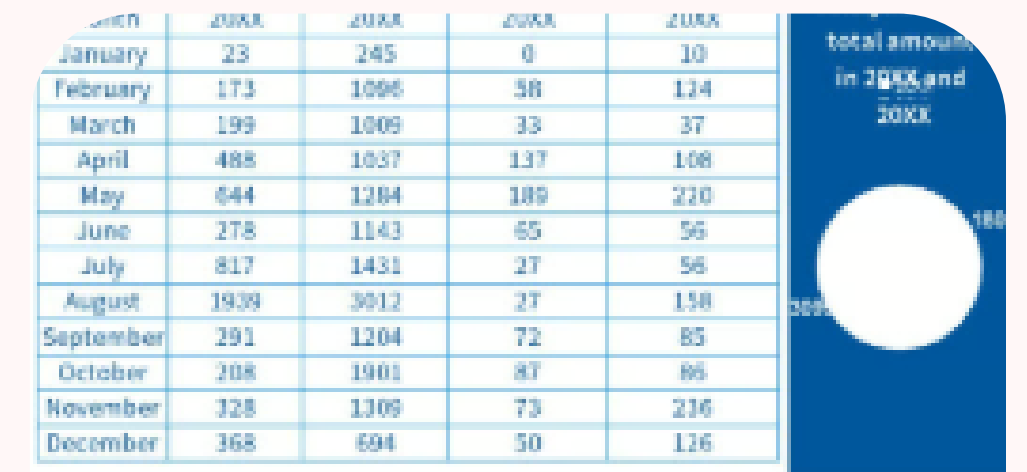
Total Features

Regions

Asia, Europe, North America

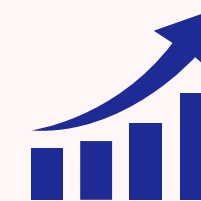
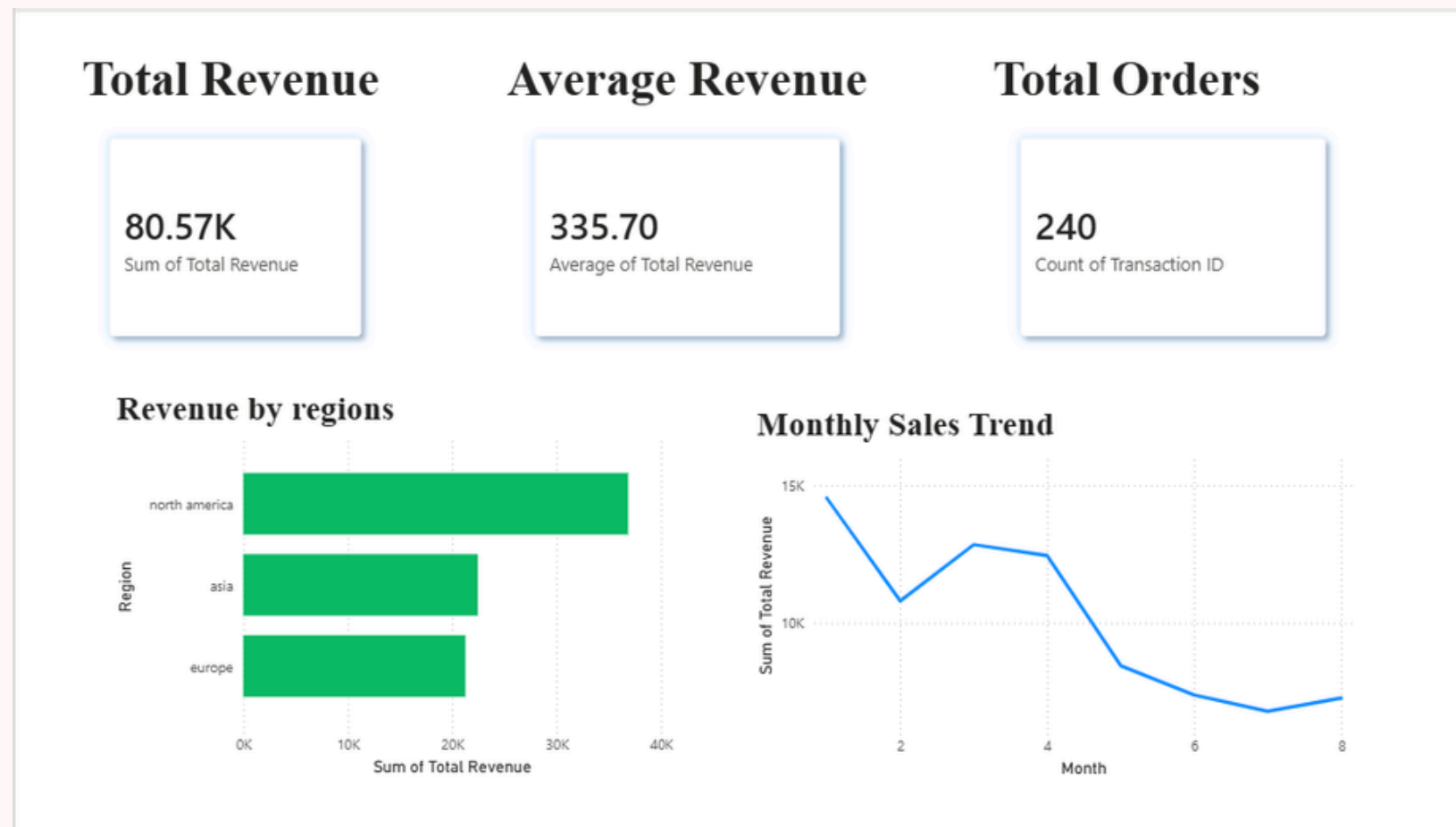
Product Categories

Electronics, Clothing, Books, Sports, Beauty Products, Home Appliances



Core KPIs

1. Total Revenue: 80.57K
2. Average Revenue per Order: 335.70
3. Total Orders: 240

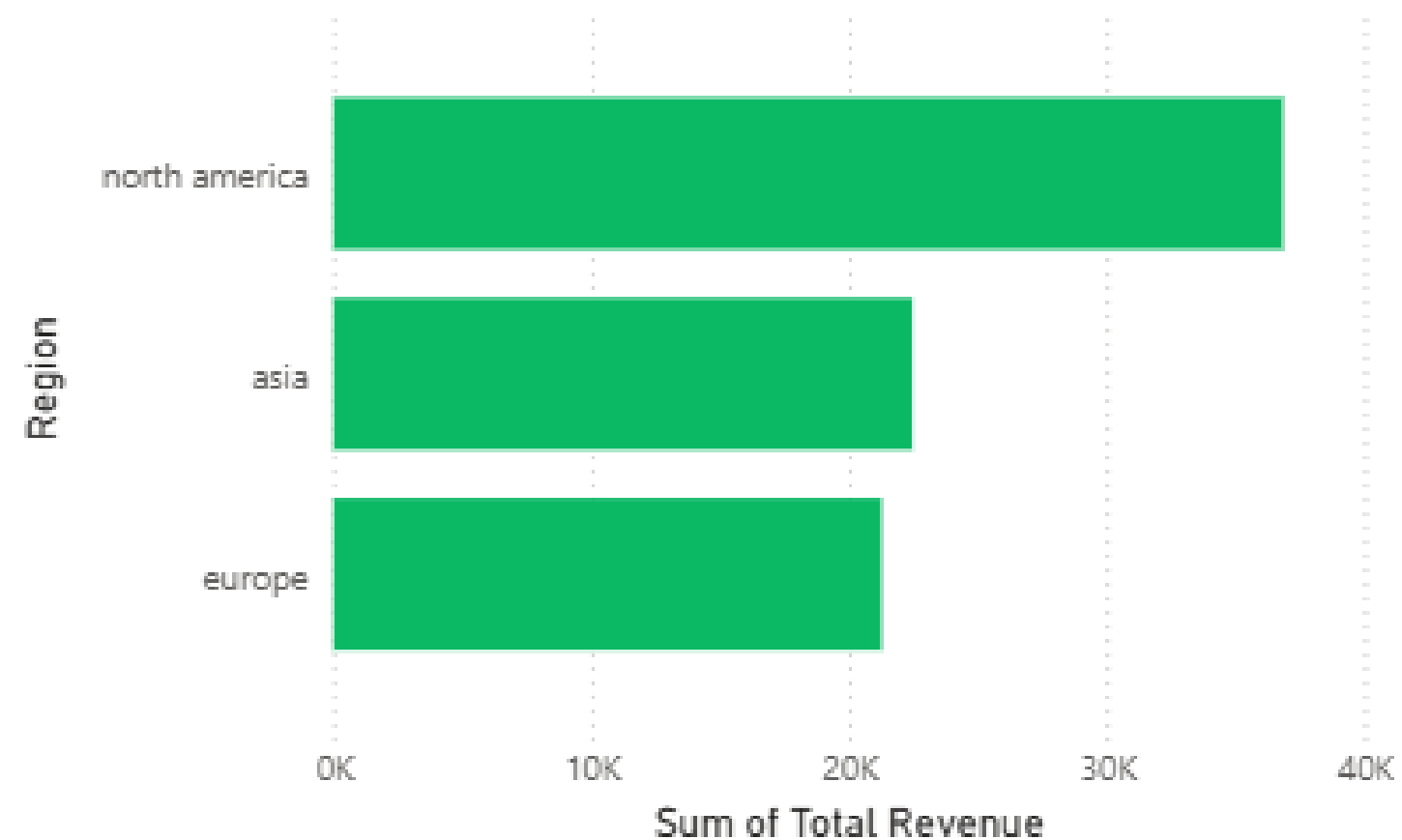


**Top Performing
Region:**
North America



**Top Product
Category:**
Electronics

Revenue by regions



Regional revenue analysis shows that North America generates the highest sales revenue compared to Europe and Asia.

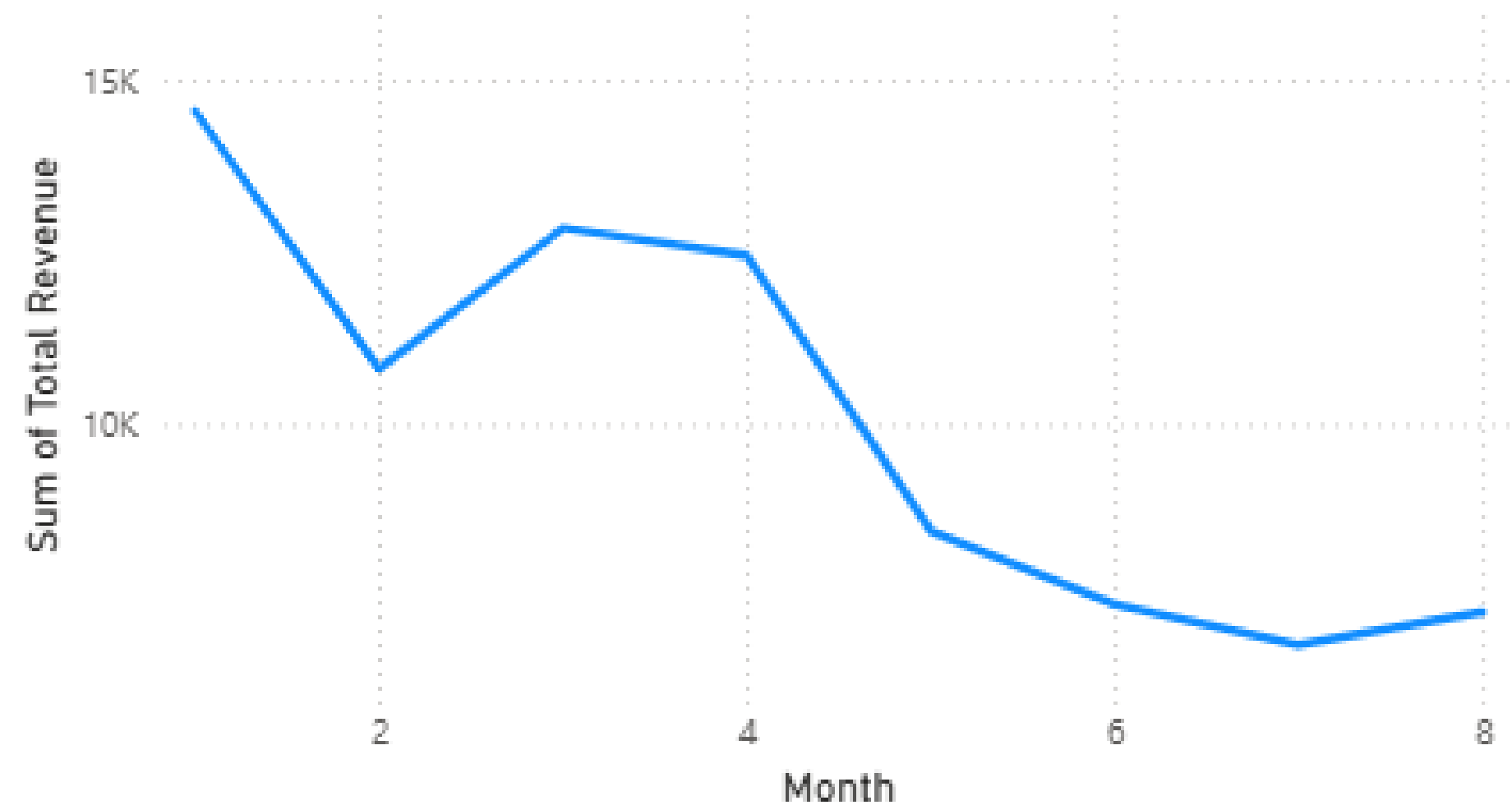
This indicates stronger customer demand and market performance in North America.

Revenue by Region

Monthly Sales Trend

The business experienced stronger revenue performance during earlier months, followed by gradual decline and stabilization.

Monthly Sales Trend

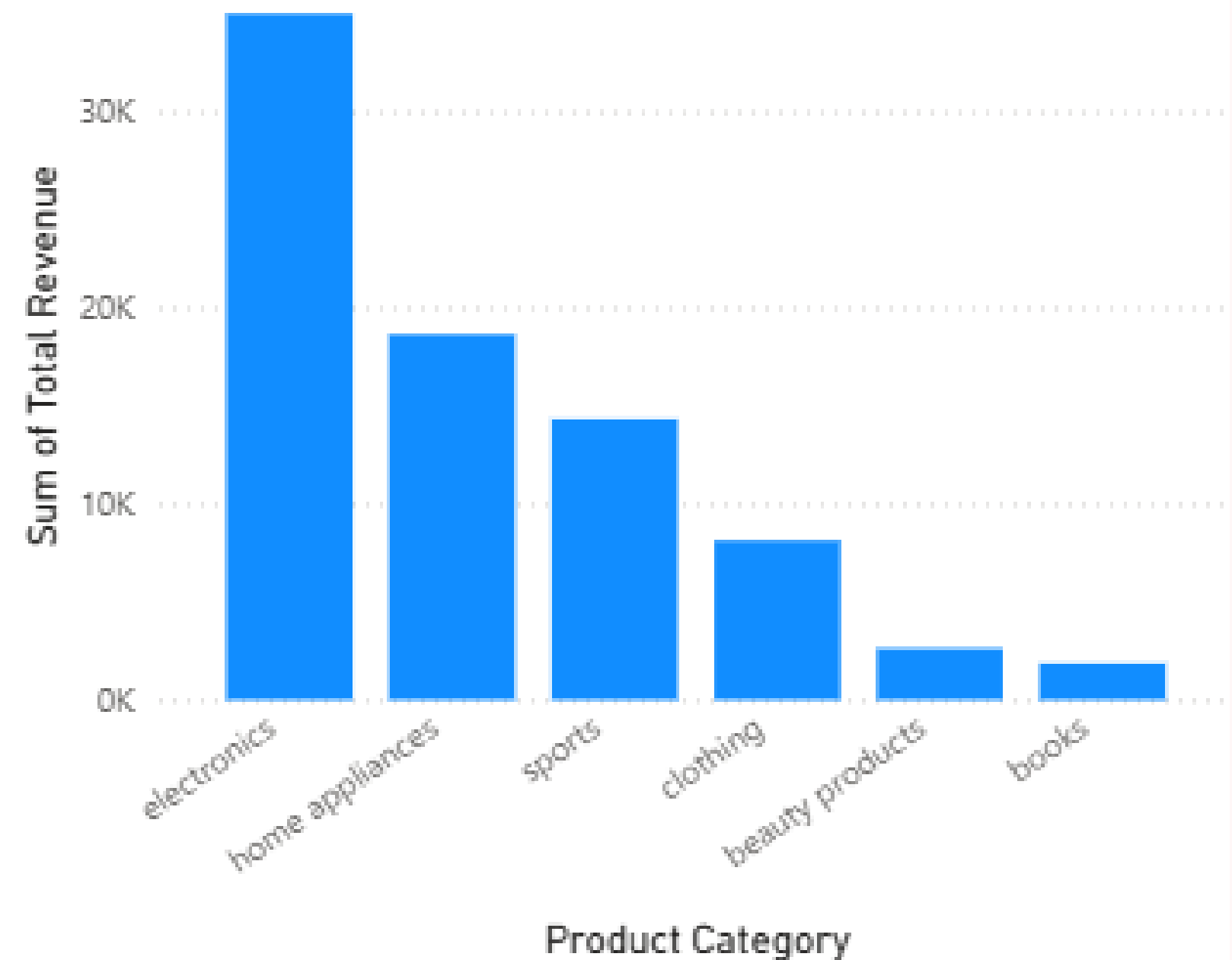


Product Category Analysis

Electronics is the highest revenue-generating category among all product categories.

Home appliances and sports products also contribute significantly to overall business revenue.

Revenue by Product Category

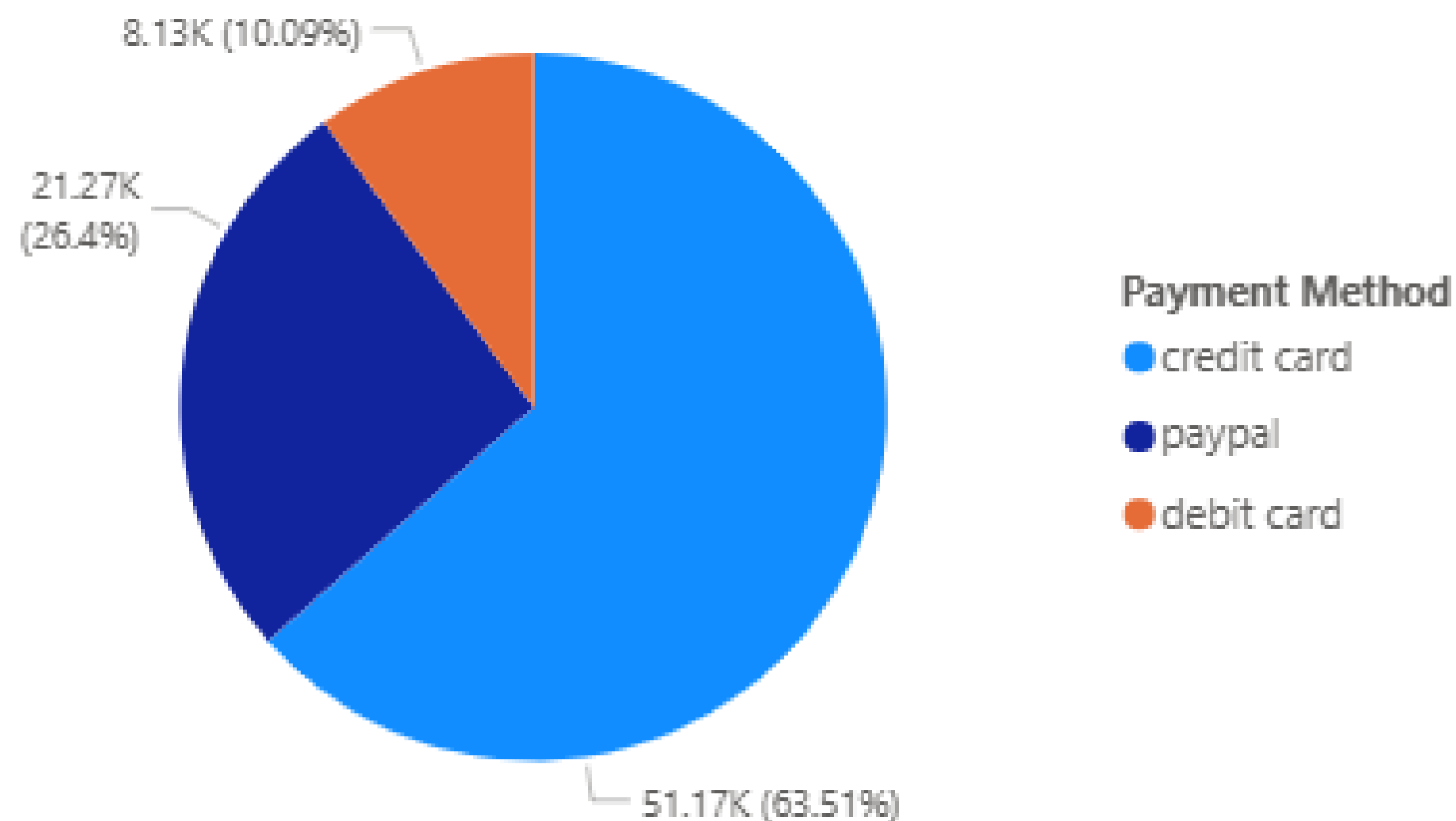


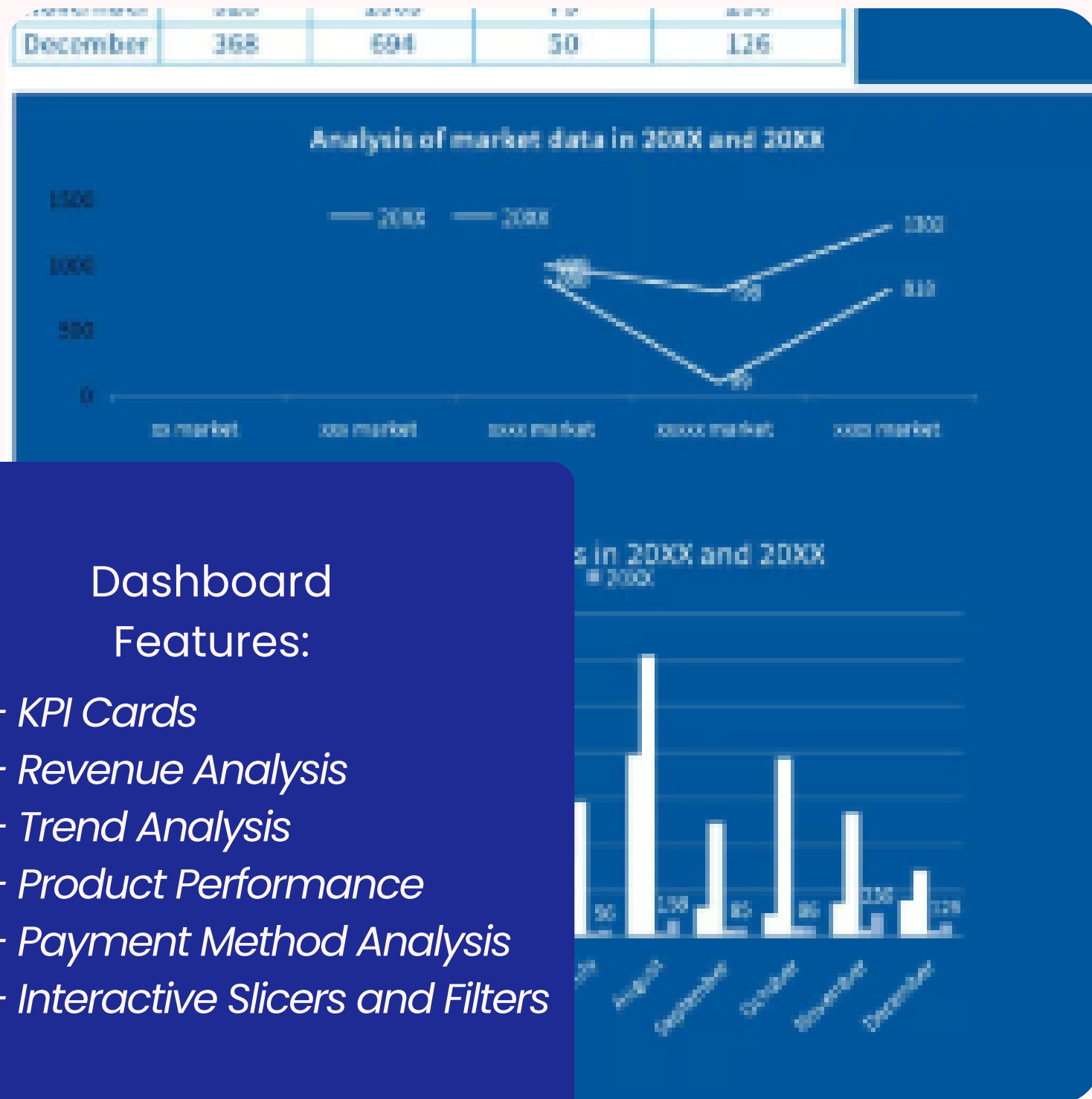
Payment Method Analysis

Customer payment analysis reveals that credit cards are the most preferred payment method.

PayPal and debit cards are used less frequently compared to credit cards.

Payment Method Analysis





Dashboard Features:

- KPI Cards
- Revenue Analysis
- Trend Analysis
- Product Performance
- Payment Method Analysis
- Interactive Slicers and Filters

Interactive Dashboard

An interactive business intelligence dashboard was created using Power BI.

Region

- ☐ asia
- ☐ europe
- ☐ north america

Product Category

- ☐ beauty products
- ☐ books
- ☐ clothing
- ☐ electronics
- ☐ home appliances

Month

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4



Hypothesis Testing

Business Question:

Does revenue significantly differ across regions?

Null Hypothesis (H_0):

There is no significant difference in revenue across regions.

Alternative Hypothesis (H_1):

There is a significant difference in revenue across regions.

ANOVA Test Results

An ANOVA test was performed to statistically compare revenue across Asia, Europe, and North America.

4.08 *F-Statistic*

This indicates that revenue differs significantly across regions, validating the business insight that some markets outperform others.

0.018 *P-Value*

Since the p-value (0.018) is less than 0.05, the null hypothesis is rejected.



Business Recommendations

- Focus marketing efforts on North America due to strong revenue performance.
- Expand electronics product offerings to maximize sales growth.
- Improve customer engagement strategies in lower-performing regions.
- Use customer payment preferences to optimize transaction systems.
- Continue monitoring regional sales trends for future business planning.

Conclusion

The project successfully combined exploratory analysis, business intelligence, storytelling, and statistical validation to uncover important sales insights.

The findings highlight strong regional and product-based revenue patterns that can support strategic business decisions and future growth opportunities.



THANK YOU!

I welcome your feedback and questions!

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